SEMESTER-VI

UCS532: COMPUTER VISION				
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	2	0	2	3.0

Course Objective: To understand the basic concepts of Computer Vision. The student must be able to apply the various concepts of Computer Vision in other application areas.

Syllabus

Digital Image Formation and low-level processing: Overview and State-of-the-art, Fundamentals of Image Formation, Transformation: Orthogonal, Euclidean, Affine, Projective, etc; Fourier Transform, Convolution and Filtering, Image Enhancement, Restoration, Histogram Processing.

Image Representation & Description: Edges - Canny, LOG, DOG; Line detectors (Hough Transform), Corners - Harris and Hessian Affine, Orientation Histogram, SIFT, SURF, HOG, GLOH, LBP and its variants, Gabor Filters and DWT.

Image Segmentation: Region Growing, Edge Based approaches to segmentation, GraphCut, Mean-Shift, MRFs, Texture Segmentation; Object detection.

Pattern Analysis: Clustering: K-Means, Fuzzy C-means; Classification: Discriminant Function, Supervised, Un-supervised, Semi-supervised; Dimensionality Reduction: PCA, LDA, ICA.

Motion Analysis: Background Subtraction and Modeling, Optical Flow, KLT, SpatioTemporal Analysis, Dynamic Stereo; Motion parameter estimation.

Self-Learning Content:

Miscellaneous: Applications: CBIR, CBVR, Activity Recognition, computational photography, Biometrics, stitching and document processing; Modern trends - superresolution; GPU, Augmented Reality; cognitive models, fusion and SR&CS.

Laboratory Work (if applicable)

To implement various techniques and algorithms studied during course.

Course Learning Objectives (CLO)

The students will be able to:

1. Understand the fundamental problems of computer vision.
2. Implement various techniques and algorithms used in computer vision.
3. Analyse and evaluate critically the building and integration of computer vision algorithms and systems.
4. Demonstrate awareness of the current key research issues in computer vision.

Text Books

1. Szeliski, R., Computer Vision: Algorithms and Applications, Springer-Verlag London Limited (2011), 1st Edition.
2. Forsyth, A., D. and Ponce, J., Computer Vision: A Modern Approach, Pearson Education (2012) 2nd Edition

Reference Books

1. Hartley, R. and Zisserman, A., Multiple View Geometry in Computer Vision Cambridge University Press (2003) 2nd Edition.
2. Fukunaga, K., Introduction to Statistical Pattern Recognition, Academic Press, Morgan Kaufmann (1990) 2nd Edition.
3. Gonzalez, C., R. and Woods, E., R. Digital Image Processing, Addison-Wesley (2018) 4 th Edition

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